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BM25MTECH14005

Assignment 1

Medical Imaging — CT Reconstruction

Q1. Sinogram of the Shepp–Logan Phantom**Background**

A **sinogram** is a two-dimensional representation of the projection data acquired during a CT scan. Each column of the sinogram corresponds to a single X-ray projection at a specific angle, while each row represents a fixed detector position. As the gantry rotates from 0° to 180° , a point source in the object traces a sinusoidal path across the sinogram—giving rise to the name.

Mathematically, the sinogram is formed via the **Radon Transform**:

$$\mathcal{R}\{f\}(r, \theta) = \int_{-\infty}^{\infty} f(r \cos \theta - t \sin \theta, r \sin \theta + t \cos \theta) dt$$

where r is the perpendicular distance from the origin to the projection line and θ is the projection angle.

Task

Generate the sinogram of a Shepp–Logan phantom using the Radon transform.

Python Code

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from skimage.data import shepp_logan_phantom
4 from skimage.transform import radon
5
6 # Generate Shepp-Logan Phantom (400 x 400)
7
8 phantom = shepp_logan_phantom()
9
10 fig, ax = plt.subplots(figsize=(5, 5))
11 ax.imshow(phantom, cmap='gray')
12 ax.set_title('Shepp-Logan Phantom')
13 ax.axis('off')
14 plt.tight_layout()
15 plt.savefig('phantom.png', dpi=150)
16
17 # Define projection angles (0 to 180 degrees, step 0.5)
```

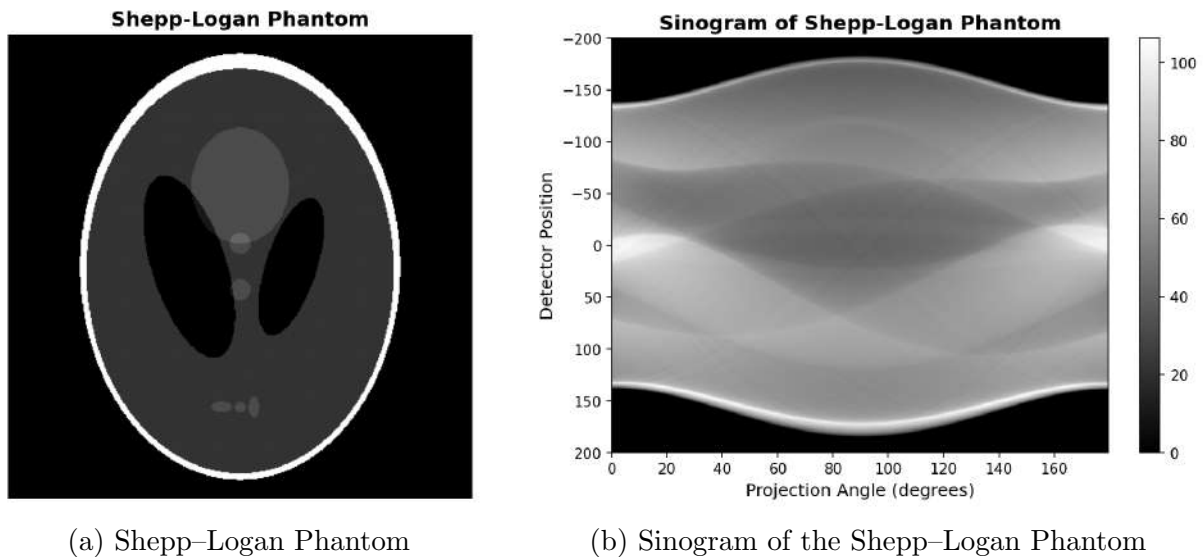
```

17 theta = np.linspace(0.0, 180.0, 360, endpoint=False)
18
19 #         Compute Radon Transform (Sinogram)
20
21 sinogram = radon(phantom, theta=theta)
22
23 #         Display Sinogram
24
25 fig, ax = plt.subplots(figsize=(7, 5))
26 im = ax.imshow(sinogram, cmap='gray', aspect='auto',
27               extent=[theta[0], theta[-1],
28                       sinogram.shape[0]//2,
29                       -sinogram.shape[0]//2])
30 ax.set_xlabel('Projection Angle (degrees)')
31 ax.set_ylabel('Detector Position')
32 ax.set_title('Sinogram of Shepp-Logan Phantom')
33 plt.colorbar(im, ax=ax)
34 plt.tight_layout()
35 plt.savefig('sinogram.png', dpi=150)

```

Listing 1: Generating the Shepp-Logan phantom and its sinogram

Results

Figure 1: Phantom image and its corresponding Radon-transform sinogram acquired over 360 projection angles (0° – 180°).

Observations

- The Shepp-Logan phantom models a simplified cross-section of a human head, consisting of nested ellipses of varying density.

- Each bright elliptical structure in the phantom produces a sinusoidal band in the sinogram, confirming the theoretical relationship between a point (or structure) and its sinusoidal trace.
- The high-intensity outer ellipse creates a wide, bright sinusoidal arc along the full angular range of the sinogram, whereas the smaller interior ellipses generate narrower, lower-intensity bands.
- Denser sampling of projection angles (finer angular step) produces a higher-resolution sinogram and subsequently a more accurate reconstruction.

Q2. CT Reconstruction via Simple Back-Projection

Background

Reconstruction from projection data is performed by inverting the Radon transform. The simplest method—**simple back-projection (SBP)**—smears each 1-D projection back across the image plane at the corresponding angle and accumulates the contributions:

$$f_{\text{BP}}(x, y) = \int_0^\pi \mathcal{R}\{f\}(x \cos \theta + y \sin \theta, \theta) d\theta$$

Because every projection contributes a uniform spread of intensity, the reconstructed image is blurred by a $1/r$ point-spread function.

Task

Simulate the step-by-step acquisition of projections and demonstrate the progression of reconstruction from sparse to dense angular sampling.

Python Code

```

1 from skimage.transform import iradon
2
3 #           Reconstruct using simple back-projection (no filter)
4 recon_bp = iradon(sinogram, theta=theta, filter_name=None)
5
6 #           Side-by-side comparison
7
8 fig, axes = plt.subplots(1, 2, figsize=(10, 5))
9 axes[0].imshow(phantom, cmap='gray'); axes[0].set_title('
    Original Phantom')
10 axes[1].imshow(recon_bp, cmap='gray')
11 axes[1].set_title('Reconstruction: Simple Back-Projection')
12 for ax in axes: ax.axis('off')
13 plt.tight_layout()
14 plt.savefig('backprojection_compare.png', dpi=150)

```

Listing 2: Simple back-projection reconstruction using `iradon`

Results

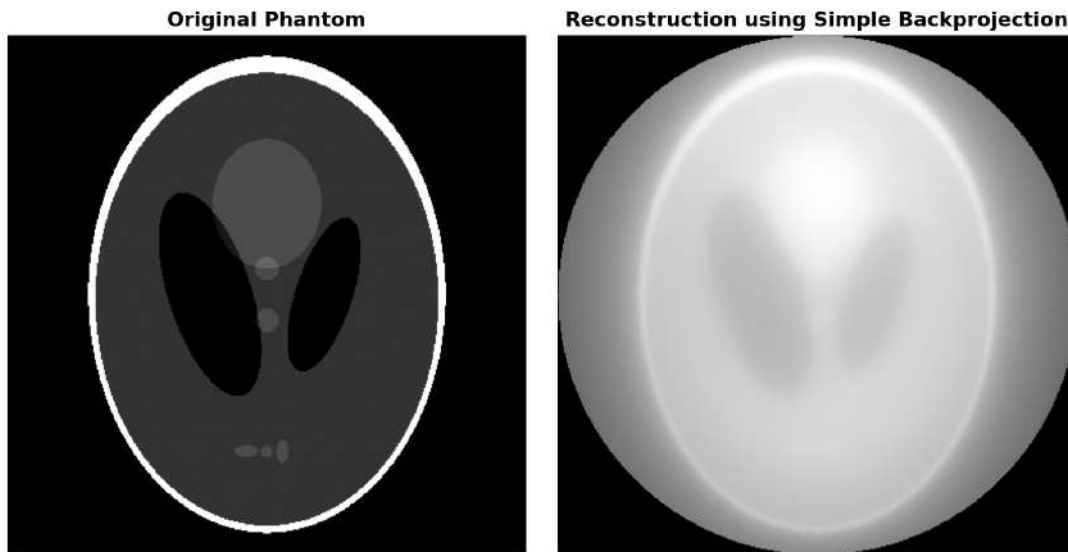


Figure 2: Left: original Shepp–Logan phantom. Right: image reconstructed via simple back-projection (no filter).

Observations and Explanation

- **Blurring effect.** The reconstructed image is visibly blurred. Back-projection distributes each 1-D projection uniformly across the reconstruction plane; summing many such smeared projections causes low-frequency components to dominate and high-frequency edge information to be suppressed.
- **Mathematical explanation.** In Fourier space, the Radon inversion requires multiplying by $|\omega|$ (the ramp filter) to compensate for the polar-to-Cartesian density of Fourier samples. Without this correction, the spectrum is weighted by $1/|\omega|$, amplifying low frequencies and producing the characteristic soft, glowing appearance.
- **Artefacts.** Star-shaped streaking artefacts radiate from high-contrast boundaries because a finite number of projections cannot perfectly cancel the back-projected wings.
- **Remedy.** Applying a high-pass (ramp-like) filter to each projection *before* back-projection—the Filtered Back-Projection (FBP) algorithm—restores edge sharpness and removes the blur.

Q3. Filtered Back-Projection with Multiple Filters

Background

Filtered Back-Projection (FBP) is the standard analytical CT reconstruction algorithm. Each projection is convolved with a 1-D filter kernel in the frequency domain before being back-projected:

$$f_{\text{FBP}}(x, y) = \int_0^\pi [\mathcal{R}\{f\}(\cdot, \theta) * h](x \cos \theta + y \sin \theta) d\theta$$

The choice of filter $H(\omega) = |\omega| \cdot W(\omega)$, where $W(\omega)$ is an apodisation window, governs the sharpness–noise trade-off.

Filters compared in this study:

Filter	Window $W(\omega)$	Characteristic
Ram–Lak (Ramp)	1	Maximum sharpness; amplifies high-freq. noise
Shepp–Logan	$\text{sinc}(\omega/2\omega_c)$	Good balance; slight noise reduction
Cosine	$\cos(\pi\omega/2\omega_c)$	Smoother, moderate noise suppression
Hamming	$0.54 + 0.46 \cos(\pi\omega/\omega_c)$	Strong noise suppression; softer edges
Hann	$0.5 + 0.5 \cos(\pi\omega/\omega_c)$	Strongest smoothing; lowest noise

Python Code

```

1 filters = [
2     ('No Filter (Backprojection)', None),
3     ('Ram-Lak Filter', 'ramp'),
4     ('Shepp-Logan Filter', 'shepp-logan'),
5     ('Cosine Filter', 'cosine'),
6     ('Hamming Filter', 'hamming'),
7     ('Hann Filter', 'hann'),
8 ]
9
10 fig, axes = plt.subplots(2, 3, figsize=(13, 9))
11 for ax, (name, filt) in zip(axes.flat, filters):
12     recon = iradon(sinogram, theta=theta, filter_name=filt)
13     ax.imshow(recon, cmap='gray')
14     ax.set_title(name, fontweight='bold')
15     ax.axis('off')
16 plt.suptitle('Filtered Back-Projection: Filter Comparison',
17             fontsize=14)
18 plt.tight_layout()
19 plt.savefig('filtered_backprojection.png', dpi=150)

```

Listing 3: Filtered back-projection with five different filter kernels

Results

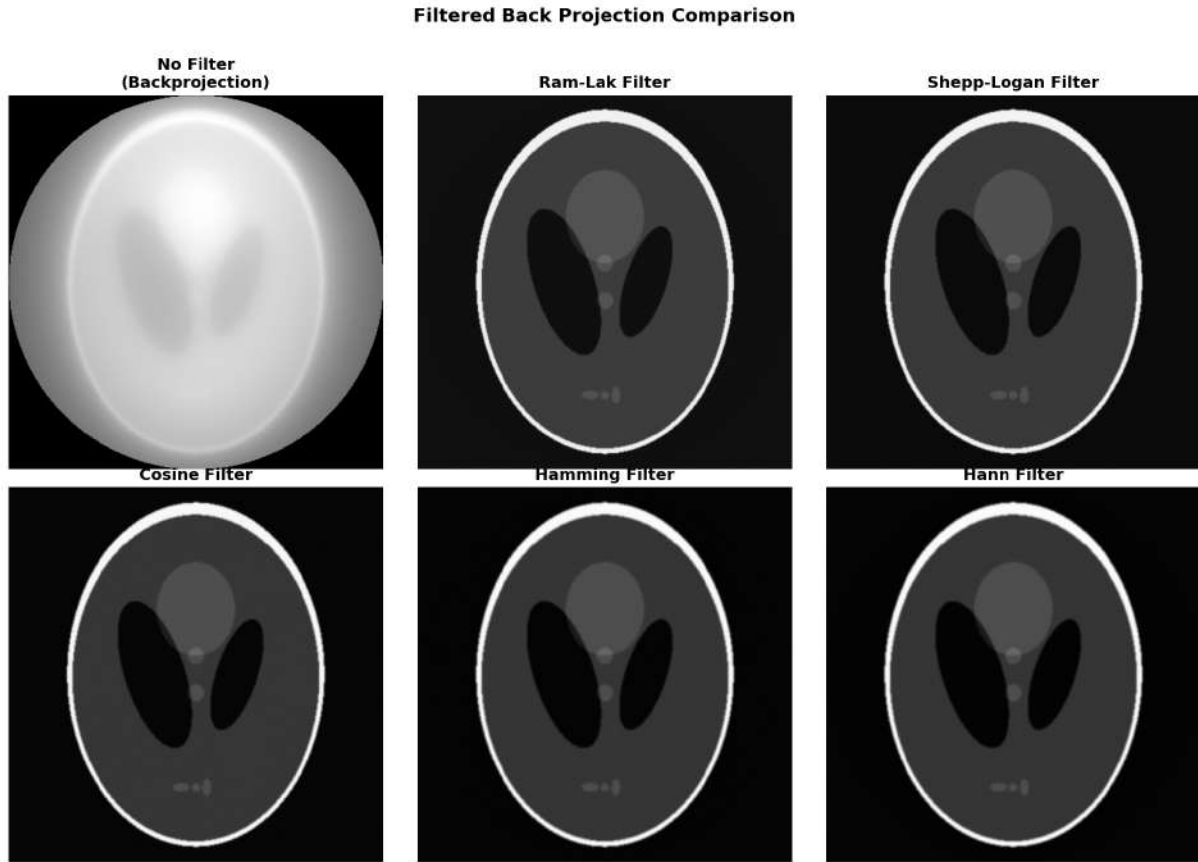


Figure 3: Side-by-side comparison of CT reconstructions using different FBP filter kernels. Top row (left to right): no filter, Ram–Lak, Shepp–Logan. Bottom row: Cosine, Hamming, Hann.

Analysis and Comparison

- **No filter.** The back-projection image is severely blurred. Structural boundaries are barely discernible and inner ellipses are completely obscured—as analysed in Q2.
- **Ram–Lak (Ramp).** The pure ramp filter delivers the sharpest reconstruction. Fine details and high-contrast edges are well preserved; however, high-frequency noise is amplified, producing a grainy appearance at low-count regions.
- **Shepp–Logan.** A sinc apodisation softens the ramp at the Nyquist limit. The result retains excellent edge fidelity with a modest reduction in noise compared with Ram–Lak—making it the default filter for clinical FBP.
- **Cosine.** A cosine window further attenuates high frequencies. Edges remain sharp but fine-grained noise is noticeably reduced; a mild low-pass blur is visible at the finest phantom features.
- **Hamming.** A raised-cosine window with a non-zero floor ($0.54 + 0.46 \cos$) provides strong noise suppression while largely retaining gross structural features. Appropriate when dose reduction (and hence higher quantum noise) is a priority.

- **Hann.** The Hann window drops to zero at ω_c , giving maximum smoothing. The image exhibits the least noise but the most blurring; fine inner details are somewhat merged.

Strategies for Further Improvement

1. **Increase angular sampling.** Acquiring more projections (smaller angular step) reduces streaking artefacts and improves the conditioning of the inverse problem, yielding sharper reconstructions even with the Hann filter.
2. **Filter selection based on clinical task.** Edge-critical tasks (e.g., lung nodule detection) favour Ram–Lak or Shepp–Logan; dose-sensitive or noise-critical tasks (e.g., low-dose paediatric CT) favour Hamming or Hann.
3. **Iterative reconstruction methods.** Algorithms such as SART, OSEM, and model-based iterative reconstruction (MBIR) incorporate physical noise models and prior knowledge, surpassing FBP in image quality at equal or lower dose.
4. **Deep-learning post-processing.** Convolutional neural networks (e.g., RED-CNN, FBPCNN) trained on paired noisy/clean sinograms can further denoise and sharpen FBP outputs without the computational cost of full iterative reconstruction.